COMPUTABILITY / BUSY BEAVER / ZFC

DR215

A direct-transfer, reparameterized consistency machine

A concrete 215-state candidate, two new local simulations, and the remaining 214-state problem

Prepared for Vladimir Reshetnikov's ProveIt project

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Abstract

We give an exact binary, one-tape Turing machine DR215 with 215 working states and an ordinary mathematical proof, formalizable in ZFC, that its blank-tape nonhalting is equivalent to Con(ZFC). The construction combines the accepted 35-occurrence single-guard transfer implementation with a changed sixth-axiom sampler. Its source has 201 distinct subtrees and 4,732 primitive occurrences. Only two transition rows differ from the accepted direct-transfer machine, but the axiom produced at a fixed command argument changes; a corresponding proof compiler is supplied.

Fresh unbounded runtime verification covers all 6,151 guarded primitive cases and all 430 transition rows. A complete 70,506-transition blank prefix matches a 177-step synthetic axiom prefix, yielding the protected startup invariant. Four additional 215-state tables realize two new predecessor-marker transformations and their compositions; their halting equivalence holds on every initial tape from state zero.

On the formalism side, antecedent exchange gives a calibrated family of numerical calculi. An exhaustive 279,936-recipe source census for one changed row attains 204 source nodes, so that experiment does not improve the state bound. All positive artifacts, exact tables, finite certificates, and portable checkers accompany the report. **No 214-state machine, new proof-assistant kernel endpoint, or state lower bound is claimed.**

Research snapshot: `b6fb232916d008383e7cf2990d68ddb30fded7db`.

Underlying numerical calculus, compiler, and machine architecture are inherited and explicitly attributed. The new contribution is the checked composition and supplementary transformations, not a new calibration of ZFC from first principles.

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1 Result and exact scope

The concrete deliverable is **DR215**, a deterministic machine with 215 working states, two tape symbols, and a bi-infinite tape. Blank is zero. The head and start state are zero, and entering a separate halt label terminates execution. The halt label is excluded from the working state count. Every working state has exactly two transition rows. The native table uses historical labels; the dense table has working labels $0, \ldots, 214$ and halt label 215.

Theorem 1.1 (Main result)**.** *For the exact DR215 table supplied with this report,*

$$\mathrm{ZFC} \vdash \quad \neg\,\mathrm{Halt}(\mathrm{DR215}, \mathbf{0}) \iff \mathrm{Con}(\mathrm{ZFC}).$$

The proof uses the repository's established numerical-calculus-to-ZFC calibration and arithmetic macro contracts. Its new source, physical simulation, and startup obligations are identified and checked below.

This is the user's 215-state alternative outcome. The 214-state target remains open in this contribution. The exact primary candidate has the following measured properties; symbolic rule applications are checker operations, not transitions of the Turing machine.

Quantity	DR215
Working states / transition rows	215 / 430
Primitive source occurrences	4,732
Distinct source subtrees	201
Guarded primitive cases	6,151
Symbolic rule applications	1,618,678
Literal transition rows covered	430 / 430
Blank startup transitions	70,506
Matching synthetic source prefix	177
Explicit startup theorem heads	10
Conservative credit needed at cutoff 1	2

The new native table has SHA-256

`fac930e9bc89fed41b631de2f12ef4e48e1c8625ef31ef093d4f14b340f113ad`.

The manifest binds the exact source, dense table, state bijection, checker code, and verification records. A source DAG count is not a physical state count: contextual decoding, arithmetic roles, and startup impose additional constraints.

1.1 What is new, and what is inherited

The direct transfer, its contextual source reassociation, the sixth-row parameter symmetry, and the general predecessor-marker idea all exist in the inspected repository. Their prior authorship is retained. The new positive results here are:

1. The exact composition of the single-guard direct transfer with the sixth-row sampler change, with its own all-natural physical verification and its own source and startup binding. It changes the actual axiom enumeration at fixed command arguments.
2. Two additional marker-role choices, $127 \mapsto 124$ and $190 \mapsto 160$, their independently checked all-tape simulations, and their compositions with each other and with the earlier $176 \mapsto 52$ transformation.

3. A calibrated antecedent-exchange family, an exact finite source census, and bounded source and physical experiments that isolate specific obstacles to a one-state reduction.

DR215 and the four supplied marker tables are distinct under rooted state renaming and global left/right reflection from all 84 distinct rooted graphs represented by 321 parseable 215-state table occurrences in the frozen comparison corpus. The corpus and its original paths are included. This proves novelty relative to that corpus, not global discovery priority and not behavioral inequivalence. The marker machines are deliberately behaviorally equivalent to their parent.

A final repository check at commit `a10b2c302cad2b3f3d0a4ce51022eb1ed5995acb` found the newly added ZC215 archive. Its main table was separately compared against all five supplied tables, again with no rooted or reflected match. The late comparison inputs and receipt are retained separately, so the earlier 84-graph corpus remains exactly reproducible.

1.2 Verification boundary

The repository's primary K215 has its own proof-assistant work. That status is not transferred to these different literal tables. DR215 has an ordinary proof and executable symbolic induction checks; no new exact Lean or Rocq endpoint was compiled for it. The checker uses Python and explicit mathematical induction principles. A complete formal endpoint would additionally formalize or import that checker soundness and bind the new literal table, source, and startup certificate.

The whole result also depends on the inherited numerical-calculus calibration. Checking a propositional recovery certificate or a runtime certificate alone does not prove the entire equivalence with ZFC. Those dependencies are stated as named reused theorems, not silently replaced by tests.

2 Numerical calculus and the changed sampler

2.1 Literal arithmetic constructors

For $a, b \in \mathbb{N}$, define

$$\mathsf{P}(a,b) = \frac{(a+b)(a+b+1)}{2} + b, \qquad \mathsf{I}(a,b) = 4\mathsf{P}(a,b), \qquad \mathsf{Q}_a b = 4\mathsf{P}(a,b) + 2.$$

Also put $\mathsf{M}(a,b) = 4\mathsf{P}(a,b) + 1$, $\mathsf{E}(a,b) = 4\mathsf{P}(a,b) + 3$, falsity $F = 1$, truth 0, and $NX = \mathsf{I}(X, 1)$. Write $X \to Y = \mathsf{I}(X, Y)$ and $\exists_a X = N\mathsf{Q}_a NX$.

All parameters and theorem codes are arbitrary natural numbers. There is no well-formedness test, support calculation, or formula parser in the machine. In particular,

$$\mathsf{I}(0,0) = 0, \qquad \mathrm{unpair}(0) = (0,0).$$

Proofs below use these as literal equalities and do not assume syntactic injectivity for expressions constructed from arbitrary numerical codes.

The inherited calculus $\mathcal{D}$ contains truth, numerical modus ponens, theorem-only generalization,

and the following schema families:

$$\begin{aligned}
R_1(a,b) &= \exists_a \mathsf{E}(a,b),\\
R_2(a,b,c) &= \mathsf{Q}_a(b \to c) \to (\mathsf{Q}_a b \to \mathsf{Q}_a c),\\
R_3(a,b,c) &= \mathsf{E}(a,b) \to (\mathsf{M}(c,a) \to \mathsf{M}(c,b)),\\
R_4(a,b,c) &= \mathsf{Q}_a \mathsf{Q}_b c \to \mathsf{Q}_b \mathsf{Q}_a c,\\
R_5(a,b,c) &= \mathsf{E}(c,a) \to (\mathsf{M}(c,b) \to \mathsf{M}(a,b)),\\
R_6(a,b,c) &= (\exists_a \mathsf{Q}_a c) \to c,\\
R_7(a,b,c) &= \mathsf{E}(c,a) \to (\mathsf{E}(c,b) \to \mathsf{Q}_{\mathsf{M}(a,b)} \mathsf{E}(a,b)),\\
L(A,B,C,U) &= ((C \to U) \to A) \to ((A \to C) \to (B \to C)), \qquad U \le A.
\end{aligned}$$

For completeness, the literal gated bundle retained by this construction is defined by

$$\begin{aligned}
B_0(b,k,m) &= \mathsf{Q}_b(m \to N\mathsf{Q}_b(m \to N\mathsf{M}(b,k))),\\
S_0(a,b,k) &= \mathsf{Q}_k(\mathsf{M}(k,a) \to \mathsf{M}(k,b)),\\
C_0(a,b,k,m) &= \mathsf{Q}_b(\mathsf{M}(b,k) \to N\mathsf{Q}_a(m \to N\mathsf{Q}_a(S_0(a,b,k) \to B_0(b,k,m)))),\\
J_0(a,b,c) &= \mathsf{Q}_c(C_0(a,b,c,c) \to NB_0(b,c,c)),\\
V_0(a,b,c) &= \mathsf{M}(a,b) \to \mathsf{Q}_c \mathsf{M}(a,b),\\
\mathrm{Bundle}(a,b,c) &= N(V_0(a,b,c) \to J_0(a,b,c)).
\end{aligned}$$

The bundle is enabled when $c \neq a$ and $c \neq b$; the generator returns truth otherwise. These literal gates and the couplings among their parameters are unchanged.

2.2 The inherited calibration

The retained dominated L schema has the same numerical theorem set as its unrestricted predecessor. The repository supplies a finite recovery certificate with 3,728 proof nodes and 5,160 term nodes; the candidate packet retains its checker and certificate. Its side conditions use $U = 0$ or a literal subterm relation that implies $U \le A$ under every natural substitution. This is a finite metaproof of schema recovery, not an extra substitution rule in the machine language.

The larger inherited calibration is

$$\mathcal{D} \vdash 1 \quad \longleftrightarrow \quad \exists p \ \mathrm{Proof}_{\mathrm{ZFC}}(p, \bot). \tag{2.1}$$

We reuse this theorem from the accepted K215 foundation and its predecessors [1, 2, 3, 4]. No weakening of the set theory bundle or replacement of ZFC by a potentially weaker theory occurs in DR215.

2.3 Changing the family enumerator

The new sixth generator is

$$R'_6(a,b,c) = (\exists_a \mathsf{Q}_a b) \to b. \tag{2.2}$$

Its third parameter is unused. Every other family remains literally unchanged. For fixed a, the old generator ranges over every matrix code via c, and the new one ranges over every matrix code via b. Consequently their sets of generated natural numbers are equal.

Lemma 2.1 (Exact calculus preservation)**.** *The calculus obtained by replacing R_6 with R'_6 has exactly the same finite derivability predicate on all natural conclusion codes as $\mathcal{D}$. Refutations translate primitively recursively in both directions.*

Proof. For an old instance with parameters (a, b, c), use new parameters (a, c, a), yielding the identical natural code. For a new instance with parameters (a, b, c), use old parameters (a, a, b). All other proof leaves and inference steps are unchanged. Replace each sixth-row leaf in a finite derivation by its identical-code counterpart. There is no restriction on any matrix code, so this argument also covers junk codes and zero aliases. □

This does *not* say that an individual AX command has the same output on the same input tuple. For example, whenever the two matrix arguments differ, the emitted sixth theorem may differ. That operational change requires the revised row-selection compiler in Section 6. Equality of schema families alone would not justify silently using the old command compiler.

3 The exact command language

A command store is (t, w, d, h): top, encoded stack, additive counter, and cached formula. Write $[x; w] = \mathsf{P}(x, w)$; several heads are nested to the right over a numerical tail. We use an *explicit* list of heads when reasoning about stack depth, since zero pairing aliases mean that numerical list length is not unique.

If $\mathrm{unpair}(w) = (x, v)$, the direct transfer is

$$\mathsf{S}(t, w, d, h) = \begin{cases} \mathsf{halt}, & h = 1, \\ (h, v, d, x), & h \neq 1. \end{cases} \tag{3.1}$$

It tests the incoming cache only. The older composite DROP;NIP;NIP also tested the incoming top; its nonhalting store effect was the same. We use the direct single-guard contract throughout.

For $y = d + x$,

$$\mathsf{MP}(x)(t, w, d, h) = \begin{cases} (y, [0; w], y, 0), & \mathsf{I}(t, y) = h, \\ (0, w, y, 0), & \mathsf{I}(t, y) \neq h. \end{cases} \tag{3.2}$$

For $\mathsf{AX}(c; x, y)$, set $U = d$, $A = d + x$, $B = h + y$, $C = c$. Let $G(A, B, C)$ be the enabled bundle or zero. The new AX result is

$$\begin{aligned} (G(A, B, C), [R_1, R_2, R_3, R_4, R_5, R_6', R_7, \\ L(A, B, C, U), \mathsf{Q}_U h, t; w], 0, 0). \end{aligned} \tag{3.3}$$

The h in $\mathsf{Q}_U h$ is the incoming cache, before adding y. The generated logical rows and gated bundle have implication tag zero; the generalization has tag two. In particular, none of those freshly constructed codes is 1. The tenth stack head saves the incoming top.

The field encodings are

$$f(\mathsf{S}) = [0], \qquad f(\mathsf{MP}(x)) = [1, x], \qquad f(\mathsf{AX}(c; x, y)) = [c + 2, y, x].$$

The retained source has logical registers $(N, T, W, A, D, H, X, Y, L)$: next candidate, top, stack, selector, counter, cache, two scratches, and residual command pointer. At a complete command boundary, $A = X = Y = 0$.

3.1 The 35-occurrence transfer body

The direct-transfer source clears T in two primitive occurrences, copies H to T while restoring H in seven occurrences, then runs the inherited 26-occurrence NIP routine. The total is $2 + 7 + 26 = 35$. NIP tests $H = 1$; on a nonhalting branch it unpairs W into the new cache and stack. Thus the full body realizes (3.1), preserves N, L, D, and returns zero selector and scratches.

All loops decrease natural registers, and the all-natural NIP contract includes $W = 0$. This body and its correct contextual association are inherited from ST215 [2]. They are not a new discovery of this packet.

3.2 Exact source and table edits

The pinned direct-transfer source is altered only at four zero-based primitive ordinals:

Ordinal	Old operand	New operand
958	register 3	register 5
963	register 3	register 5
1136	register 3	register 5
1141	register 3	register 5

These are the two decrement-and-restore copies of the sixth row's matrix. Register 3 holds C and register 5 holds B in that AX context. Changing both references in each copy preserves the copied register and changes the selected matrix exactly as in (2.2). Every opcode, source address, and guarded successor ordinal remains unchanged. The source remains at 201 distinct subtrees, 4,732 occurrences, depth 44, and maximum zero run 6.

The complete native transition delta from ST215 is

State/read	Old action	New action
199/1	1 R 166	1 R 12
206/0	0 R 186	0 R 76

No state is added or removed. The exact reconstruction uses pinned parent files and produces both native and dense tables; it does not invoke a solver. The source is checked using two independent control-flow resolvers. These finite checks establish literal identities; the physical runtime check below establishes their unbounded operational meaning.

4 Unbounded physical simulation

A standard tape frame has dispatcher head at integer position p, a marker at $p-1$, source-address bits beginning at p, then a zero gap to a header at $p+64+g$, where $g \in \mathbb{N}$. Register blocks are

$$B(v) = 0\,1^{v+1}.$$

Logical registers map to physical block indices

$$(N, T, W, A, D, H, X, Y, L) \longmapsto (6, 2, 0, 4, 3, 7, 8, 5, 1).$$

After nine primary blocks is a nonempty finite spare list, whose values are arbitrary naturals. The rest of the tape is zero. The represented frame includes the entire tape, not only decoded register values.

For every primitive occurrence, decrements split into zero and positive cases. The supplied runtime derivation has 6,151 cases and quantifies over arbitrary primary register values, gap, first spare, and remaining spare list. It uses three kinds of sound inference:

1. A literal transition from the exact table, with its precise write, head motion, control target, and clock increment.
2. Natural-number induction over a unary run scanned by a verified directional self-loop.
3. Structural induction over an arbitrary block list, after deriving the required one-block action with opaque left and right contexts.

The four block schemas scan B right, convert B to $C = 1\,0\,1^v$ right, scan B left, and convert C back to B left. Each has duration $v + 2$. List normalization uses the inductively justified identity $C(A)\,1 = 1\,B(A)$. The checker retains all unaffected tape contexts.

Affine expressions have a constant coordinate and twelve nonnegative variable coordinates: gap, nine primary values, first spare, and width of the remaining spare list. Each successful case has a finite positive duration. A positive constant and nonnegative variable coefficients prevent an infinite source computation from being simulated in finite binary time. Increment adds a zero spare; no primitive deletes the last spare, so the frame class is closed.

For DR215, the fresh run completed all 6,151 cases with no failures, 1,618,678 symbolic rule applications, and coverage of all 430 literal rows. The exact input identities and per-case records are supplied. This is an unbounded symbolic induction argument, not a test suite over small integers. It nevertheless remains an ordinary checker-backed proof with the explicitly stated implementation boundary.

Proposition 4.1 (Physical correspondence)**.** *Each source primitive from a standard frame is simulated by a finite positive number of DR215 transitions, returning the complete frame for its literal guarded source successor, or entering the separate halt label exactly for the halting primitive. Consequently finite halting and infinite source execution lift to the corresponding binary behavior.*

Proof. Apply the exact-table symbolic derivation to the appropriate occurrence and guard, then induct over source steps. Frame closure justifies each successive application. Positivity of durations supplies divergence preservation. The unique halt case is checked against the literal halting row. □

5 Exact startup and why the opaque tail is safe

The complete native blank execution reaches the following frame after 70,506 transitions:

$$q = 84, \qquad p = -2, \qquad \text{header} = 62, \qquad g = 0,$$
$$\text{PC} = 00110010010110010, \qquad \text{ordinal} = 58,$$
$$(N, T, W, A, D, H, X, Y, L) = (1, 0, 0, 0, 0, 0, 0, 18, 0).$$

There are 63 zero-valued spare blocks and exactly 100 one-cells. The certificate records the complete one-cell set; every other tape cell is zero. Dense-table replay uses the supplied bijection and agrees with the native run. The altered transition rows do not occur in this finite startup prefix, but startup is still checked on the new table.

This is an interior AX frame, not a clean command-boundary store. Starting the source at AX entry ordinal 25 with

$$(N, T, W, A, D, H, X, Y, L) = (1, 0, 2, 0, 0, 0, 0, 0, 0)$$

reaches the same ordinal and the entire same logical store after exactly 177 primitive steps. Both source-flow resolvers agree. The input is $\mathsf{AX}(0; 0, 0)$ on $(t, w, d, h) = (0, 2, 0, 0)$; at those parameters the old and new sixth-row matrices coincide.

The all-natural AX contract therefore applies to the remaining suffix. It terminates at a command boundary with $N = 1$, $L = A = X = Y = 0$, $t = d = h = 0$, and

$$w = [R_1(0, 0, 0), \ldots, R'_6(0, 0, 0), R_7(0, 0, 0), L(0, 0, 0, 0), \mathsf{Q}_0 0, 0; 2].$$

These ten explicit heads are theorems. The bundle gate is zero at the all-zero arguments. No literal simulation of the enormous completed formula codes is needed: exact matching of a finite prefix, terminating AX semantics, and the universal primitive simulation determine the suffix.

The numerical tail 2 is not harmless on its own:

$$2 = \mathsf{P}(0,1), \qquad 1 = \mathsf{P}(1,0).$$

Repeated unprotected pops would expose falsity. The proof below shows that the actual schedule never reaches this tail. Historical bootstrap examples in inherited documents that use ordinal 55 or tail 4 belong to different tables; the displayed frame and prefix are the DR215 bindings.

6 The revised strong proof compiler

For every numerical theorem X, we construct a finite block B_X with the following contract for *all* natural t, w:

$$(t, w, 0, 0) \xrightarrow{B_X} \mathsf{halt} \quad \text{or} \quad (X, w, 0, 0). \qquad \text{(Strong)}$$

The stack need not contain theorems. Early halting is allowed for this completeness contract; the soundness argument separately establishes theorem-valued actual execution.

6.1 Stack operations without an extra opcode

Put $D = \mathsf{AX}(0;0,0)$ and define

$$H = D;\mathsf{S}^{11}, \qquad \mathrm{PUSH} = D;\mathsf{S}^{9};\mathsf{MP}(0).$$

After ten transfers following D, the store is $(\mathsf{Q}_d h, w, 0, t)$. The first ten guards see only zero or fresh nonfalsity heads. The eleventh transfer halts exactly when the original $t = 1$; otherwise it preserves t, pops one original stack head into cache, and resets the counter. Hence for $w = [x; v]$,

$$H(t,w,d,h) = \mathsf{halt}\ (t = 1), \quad H(t,w,d,h) = (t, v, 0, x)\ (t \neq 1).$$

In PUSH, nine transfers instead leave the generalized cache, of tag two, as the tested MP cache. The implication comparison has tag zero, so MP must fail. Therefore PUSH always returns $(0, [t; w], 0, 0)$ and never halts.

For zero incoming counter and cache, define

$$\mathrm{DISCARD} = \mathsf{MP}(1);\mathsf{AX}(1;0,0);\mathsf{S}^{10};\mathsf{MP}(0).$$

The first MP fails, yields counter one and top zero, and preserves the stack. AX has $A = C = 1$, hence gate zero. Ten transfers expose a quantifier code while preserving the old stack and zero cache. The last MP fails and returns $(0, w, 0, 0)$. This gives the truth block B_0.

When top, counter and cache are zero, let $J(0)$ be empty and let $J(u) = \mathsf{MP}(u)$ for $u > 0$. It sets the counter to u without disturbing the stack. The empty zero case is essential: $\mathsf{MP}(0)$ at all-zero comparison data would succeed by $\mathsf{I}(0,0) = 0$ and push another zero.

6.2 The repaired sixth-row selector

For an unchanged retained row $R_j(a, b, c)$, use

$$\mathrm{DISCARD};\mathsf{AX}(c;a,b);\mathsf{S}^{j+1};H^{9-j}.$$

The transfers place the selected row on top. Repeated H then removes the remaining generated entries and saved zero while retaining that nonfalsity top; the original stack and zero counter/cache are restored.

For a desired old sixth-row theorem with binder a and matrix m, the new selector is instead

$$\boxed{\mathrm{DISCARD};\mathsf{AX}(a;a,m);\mathsf{S}^7;H^3.} \tag{6.1}$$

After DISCARD, the AX arguments give $A=a$, $B=m$, $C=a$, $U=0$. The sixth head is exactly $(\exists_a\mathsf{Q}_a m)\to m$ and its gate is zero. The same stack-count calculation proves (Strong). This explicitly repairs the operational effect of changing the sampler.

For $L(a,b,c,u)$ with $u\le a$, use

$$\mathrm{DISCARD};J(u);\mathsf{AX}(c;a-u,b);\mathsf{S}^9;H.$$

For an enabled bundle, use $\mathrm{DISCARD};\mathsf{AX}(c;a,b);H^{10}$. These formulas preserve the original gates and dominance constraint.

6.3 Generalization and modus ponens

Given a compiled theorem X, generalization is implemented by

$$B_{\mathsf{Q}_i X}=B_X;\mathrm{PUSH};J(i);\mathsf{S};\mathsf{AX}(i;0,0);\mathsf{S}^{10}.$$

The single transfer loads X into cache without testing the newly loaded value. AX then puts Q_iX in its ninth head, has zero gate because $A=C=i$, and saves zero. Ten transfers restore the original stack. This calculation also covers $i=0$ and $X=1$.

For major premise $M=\mathsf{I}(A,B)$, build the major before the minor:

$$\begin{aligned}B_B=B_M;\mathrm{PUSH};B_A;\mathrm{PUSH};\\ \mathsf{S};\mathsf{S};\mathsf{MP}(B);\mathsf{AX}(B;0,0);\mathsf{S}^{11}.\end{aligned}$$

After the first four blocks the store is $(0,[A,M;w],0,0)$ unless an allowed halt has occurred. Two transfers give $(A,w,0,M)$, or halt already if $A=1$. MP succeeds by the literal equation $M=\mathsf{I}(A,B)$, giving $(B,[0;w],B,0)$. The following AX and eleven transfers remove exactly that additional zero and restore $(B,w,0,0)$, or halt when $B=1$. For $A=B=0$, MP intentionally succeeds at the native zero alias; the same final transfer removes its extra stack entry.

Induction on a finite derivation proves (Strong), with the sixth-row case handled by (6.1). A finite proof DAG can be expanded to a tree; major-before-minor order introduces no logical dependency. In particular, if $\mathcal{D}\vdash 1$, then $B_1;H$ halts from every store $(t,w,0,0)$.

6.4 Cofinal successful candidate numbers

Let $F(fs,r)$ right-fold a field list by pairing over r, and define

$$\mathrm{tail}([])=0,\qquad \mathrm{tail}(k::L)=1+F(f(k),\mathrm{tail}(L)).$$

The successor belongs to continuation encoding because a positive residual is decremented before its next command. A fresh candidate is decoded directly, without that outer successor.

Set $R=\mathrm{tail}(B_1;H)$. For every $Y\in\mathbb{N}$, the fresh candidate

$$N_Y=\mathsf{P}(2,\mathsf{P}(Y,\mathsf{P}(1,R))) \tag{6.2}$$

starts with $\mathsf{AX}(0;1,Y)$, resetting counter and cache from any retained store. Its suffix therefore halts by (Strong). Since $N_Y\ge Y$, successful candidates are unbounded.

Every positive residual strictly decreases after command decoding: it becomes a right projection of its predecessor after the appropriate number of unpairings. Each arithmetic command terminates on natural registers. Thus each nonhalting candidate drains, and the unchanged enumerator, which increments the candidate number by one, reaches a chosen successful N_Y unless it has already halted.

7 Protected credit for the real schedule

The following argument uses the actual field arities and candidate order. It is independent of the values of the axiom matrices, so the sampler change does not alter it. Let f be the right Cantor projection, $T_s = s(s+1)/2$, and $W(n)$ the syntactic command word of fresh candidate n, ignoring possible semantic halts. Put $R(0) = []$ and $R(r) = W(r-1)$ for positive r. If $\mathrm{unpair}(n) = (a, y)$,

$$W(n) = \begin{cases} \mathsf{S}; R(y), & a = 0, \\ \mathsf{MP}; R(f(y)), & a = 1, \\ \mathsf{AX}; R(f(f(y))), & a \geq 2. \end{cases} \tag{7.1}$$

Assign conservative weights $\mathsf{S} = -1$, $\mathsf{MP} = 0$, $\mathsf{AX} = 10$; successful MP's extra truth head is omitted from the lower bound. Let $b(n)$ be the word's total weight, $B(k) = \sum_{n<k} b(n)$, $g(0) = 0$, and $g(r) = b(r-1)$ for $r > 0$. Induction gives $|W(n)| \leq n + 1$, so every prefix of $R(r)$ loses at most r units.

Lemma 7.1 (Prefix pullback)**.** *If a sequence has nonnegative initial partial sums, composing it with f or with f^2 preserves that property.*

Proof. In argument order, the right Cantor projection concatenates $[0], [0,1], [0,1,2], \ldots$. Every finite prefix consists of complete lists and an initial part of the next list. Each corresponding sum is nonnegative. Apply the same argument again for f^2. □

Lemma 7.2 (Credit bound at cutoff one)**.** *The minimum cumulative conservative weight over every command prefix of $W(1)W(2)W(3)\cdots$ is -2.*

Proof. The initial words are $W(0) = \mathsf{S}$, $W(1) = \mathsf{MP}$, $W(2) = \mathsf{SS}$, so $B(0), \ldots, B(3) = 0, -1, -1, -3$. Jointly induct on Cantor diagonals to show $B(k) + k \geq 0$ and that each diagonal $s \geq 2$ is prefix-safe relative to its start.

Use $q(r) = g(r) + 1$. For $k > 0$ its prefix sum is $k + B(k-1) \geq 0$. On diagonal $s \geq 2$, the first $s-1$ candidates start with AX, followed by MP and then S. The twice-pulled-back inequality gives at least $9k$ weight from the first k complete AX candidates. Before AX number y the balance is at least $9y$; after its initial $+10$, its continuation loses at most $f^2(y) \leq y$. Every AX prefix is therefore safe.

After the AX portion the balance is at least $9(s-1)$. The MP suffix can lose at most $s-1$ and the final S word at most $1 + s$. Hence prefixes through the last two words are bounded below by

$$9(s-1) - (s-1) - 1 - s = 7s - 9 > 0.$$

All requested pullback indices lie on earlier diagonals. Combining diagonal safety with $B(T_s) \geq -3$ and $k \geq 3$ preserves $B(k) + k \geq 0$, closing the joint induction.

Starting at cutoff one, the exceptional prefix is MPSS, of minimum weight -2, and no later diagonal reduces the balance. This establishes both the upper need and its attainment for the conservative syntactic weights. □

Maintain an explicit finite sequence of theorem heads above a fixed opaque tail J. AX adds ten theorem heads, successful MP adds truth, failed MP adds none, and S consumes one head. Top and cache remain theorems whenever no halt has occurred. Starting with ten heads and debt at most two leaves at least eight explicit heads above J at every boundary. Before a transfer there is therefore a head to consume; the tail is never exposed. This uses only exact inversion of $\mathsf{P}(x, v)$ and does not assume unique encoded list length.

DR215 starts ordinary enumeration at candidate one with those ten heads over $J = 2$. Its sole halting test can therefore succeed only when the cache is a theorem equal to 1.

8 Proof of the consistency equivalence

Proof of Theorem 1.1. The exact blank prefix reaches the displayed interior AX frame. Its matching source prefix and terminating AX contract complete it to ten theorem heads over tail 2, with next candidate one and empty residual. Universal physical correspondence transfers this conclusion to the literal binary table.

If DR215 halts, the protected-credit invariant and command equations show that its cache is the numerical theorem 1. Exact calculus preservation and (2.1) then give a finite ZFC refutation.

Conversely, a finite ZFC refutation yields $\mathcal{D} \vdash 1$ by the inherited calibration. The revised compiler produces the universal halting word $B_1; H$. The cofinal shell (6.2) gives a successful fresh candidate above the actual cutoff, regardless of the retained stack. Every preceding nonhalting candidate terminates, so the enumerator reaches it or halts earlier. Its finite source halting lifts to finite binary halting.

The resulting equivalence

$$\mathrm{Halt}(\mathrm{DR215}, \mathbf{0}) \longleftrightarrow \exists p \ \ \mathrm{Proof}_{\mathrm{ZFC}}(p, \bot)$$

uses finite calculations, primitive-recursive translations, and induction on natural numbers, finite lists, and finite derivations. These arguments are formalizable in ZFC without assuming consistency. Negating the equivalence proves the theorem. □

No unconditional nonhalting claim is made. If ZFC is consistent, the machine does not halt and the usual second-incompleteness argument prevents ZFC from proving this nonhalting statement. Mere consistency is not being used to assert that ZFC cannot prove a false halting statement.

9 Two new predecessor-marker transformations

The supplementary family starts from the primary K215 native table, not from DR215. Its parent SHA-256 is

`9683d6ed9628e6ce407c315b08f5da3d063da94e92706f275ca0f33e8272e4b2`.

Each transformation merges two logical boundary roles and reuses the freed physical label as an internal selector. Consequently the state count stays at 215. The gain is a different physical organization with a simple all-tape simulation, potentially useful for later sharing.

9.1 M124: distinguish old roles 124 and 127

The old states have a common read-one action $1R138$, while their read-zero actions are $0R62$ and $0R14$. The only incoming row to 124 is $104/1 : 1R124$, and the only incoming row to 127 is $110/0 : 0R127$. Therefore, on every entry from start zero, the cell immediately left of the head records the role: one for 124 and zero for 127, on every initial tape.

Map old role 127 to boundary role 124 and reuse physical state 127 as the selector. Exactly four rows change:

Row	Old	New
$110/0$	$0R127$	$0R124$
$124/0$	$0R62$	$0L127$
$127/0$	$0R14$	$0R8$
$127/1$	$1R138$	$1R59$

Provider rows $8/0 : 0R14$ and $59/0 : 0R62$ complete the simulated step. On current zero the new machine goes left to inspect the unchanged marker, right to the selected provider, then right to the old target. Both cells retain their original symbols. On current one, the common old action executes directly.

9.2 M160: distinguish old roles 160 and 190

Here the common action is read-zero $0R99$, while the read-one actions are $1R99$ and $1R52$. The only incoming rows are $139/0 : 0R160$ and $165/1 : 1R190$, so the predecessor marker is zero for old 160 and one for old 190. Map $190 \mapsto 160$ and change:

Row	Old	New
$160/1$	$1R99$	$1L190$
$165/1$	$1R190$	$1R160$
$190/0$	$0R99$	$0R86$
$190/1$	$1R52$	$1R193$

Provider rows $86/1 : 1R99$ and $193/1 : 1R52$ give the same exact three-step simulation, this time on exceptional current symbol one.

9.3 Composition and global proof

The M124 and M160 changed rows are disjoint, and neither modifies a provider row used by the other. They compose into an eight-row edit. They also compose with the repository's earlier $176 \mapsto 52$ conditional-marker transformation, producing a fourteen-row edit. For that inherited transformation the exceptional incoming route $189/1; 156/1$ forces the current cell to one and cannot trigger its read-zero discriminator; the other incoming routes write the required zero predecessor. The builder and independent auditor check the entire incoming-edge enumeration.

Theorem 9.1 (All-tape halting equivalence)**.** *For each of the four supplied marker variants, starting old and new machines in state zero at the same head position on any binary tape, the machines halt together. There is a checkpoint map with exact whole tape and head equality, control mapped by the stated role identifications, and times $t(n)$ satisfying $n \le t(n) \le 3n$.*

Proof. Induct on old steps. A nonexceptional row is a literal copy with targets mapped by the role map, so it uses one new step. An exceptional row uses the verified three-step marker/provider calculation. Incoming-edge invariants establish the required marker at every such boundary. The independent check compares both local cells and confirms that the block reads nothing outside them, leaving arbitrary exterior tape intact.

Set $t(n+1) - t(n)$ to one or three accordingly. Internal selector and provider states are working states, so no exceptional block halts inside. A halting transition occurs at corresponding boundaries. The bound $t(n) \ge n$ rules out a finite new run representing infinitely many old steps, yielding the converse as well. □

This proves the same ZFC consistency equivalence as the accepted parent. It is stronger than a blank-tape simulation, but makes no claim about starting arbitrarily inside a replaced role. Finite startup checks are corroboration rather than the basis of the all-tape proof: M124 and its compositions reach the old 95-one-cell seed after 95,840 transitions; M160 reaches it after 95,714. Every mapped old-step boundary and the complete final tape were compared.

10 A calibrated family of changed formalisms

The user's permission to change the formalism is substantial: only refutability, not theorem-for-theorem identity, ultimately needs to be preserved. The positive formalism experiments here establish the stronger identity property, because it supplies a short trustworthy calibration while allowing a different axiom layout.

10.1 Antecedent exchange on numerical codes

Retain the dominated L basis, which recovers the unrestricted basis and hence the propositional K and S schemata. They derive

$$(X \to (Y \to Z)) \to (Y \to (X \to Z)). \tag{10.1}$$

For an explicit derivation template, assume $f : X \to (Y \to Z)$, $y : Y$, and $x : X$. Two MP steps yield Z. The finite K/S deduction construction discharges x, then y, then f. It replaces the discharged assumption by $A \to A$, any other assumption or theorem by a K instance, and an MP step by an S instance. No GEN is used, so no generalization side condition is hidden. Exchange X, Y for the reverse implication.

Choose any subset of the following four rows and exchange their two curried antecedents:

Row	X	Y	Z
R_2	$\mathsf{Q}_A(B \to C)$	$\mathsf{Q}_A B$	$\mathsf{Q}_A C$
R_3	$\mathsf{E}(A, B)$	$\mathsf{M}(C, A)$	$\mathsf{M}(C, B)$
R_5	$\mathsf{E}(C, A)$	$\mathsf{M}(C, B)$	$\mathsf{M}(A, B)$
R_7	$\mathsf{E}(C, A)$	$\mathsf{E}(C, B)$	$\mathsf{Q}_{\mathsf{M}(A,B)}\mathsf{E}(A, B)$

Both old and changed calculi retain the same K/S basis. Each derives every missing axiom from its partner by (10.1) and one MP. Replacing axiom leaves gives primitive-recursive translations of all finite proofs, preserving every conclusion code, including truth and falsity. The argument applies at arbitrary natural parameter values and requires no formula parser or ordinary syntactic substitution theorem.

Independent permutations of the three parameter positions in a row also preserve its full schema family. The command compiler selects an instance by resetting counter/cache and using the inverse permutation for the AX fields, as the sixth-row example demonstrates. All gates and private parameter couplings in the set theory bundle remain unchanged.

10.2 Measured source cost

In the primary K215 source, the merged equality row R_7, called G by the source tools, occupies primitive interval $[479, 888)$. Its first two atom constructors occupy $[479, 643)$. Swapping registers four and five in the CI/CD operands of that prefix exchanges exactly those two antecedents, without changing any source address, opcode, or guarded successor. The resulting source has 205 distinct subtrees, against the primary baseline's 202.

An exhaustive census then applies all $6^7 = 279{,}936$ independent parameter permutations of the seven logical rows to that exchanged source. At the retained literal association, the minimum is 204 source nodes, achieved by four recipes. All recipes keep 4,751 primitive occurrences. A subsequent width-64 closed-reassociation beam visited 76,675 sources over eleven rounds and saved no source below 204. These counts are scoped to those finite or bounded search families.

We also reran both finite families on DR215 itself. All 279,936 row permutations have minimum 201, with four minimizers. After antecedent exchange in R_7, the same exhaustive census has

minimum 203, again with four minimizers. A width-64 closed-reassociation beam on DR215 visited 75,077 sources over eleven rounds and retained minimum 201. The portable replays reproduced the histograms and every minimizing source byte string. Thus no source of 200 or fewer nodes was found in these precisely specified families.

Single changed row / construction	Best saved DAG	Zero run
Exchanged R_2, closed-row re-emission	205	6
Exchanged R_3, closed-row re-emission	205	6
Exchanged R_5, closed-row re-emission	207	6
Exchanged R_7, closed-row re-emission	206	7
Exchanged R_7, original association	205	6
Dual R_6, experimental and uncalibrated	203	8

The last row experiments with $C \to \mathsf{Q}_A \exists_A C$ in place of $(\exists_A \mathsf{Q}_A C) \to C$. Its apparently familiar classical duality does not establish the needed result: the usual derivation uses ex falso, while the removed original row itself supplies ex falso in this numerical basis. The repository already contains a countermodel to equality of the two theorem sets. A separate relative refutation translation could still be useful, but none is proved here. That row is not an accepted candidate and contributes no machine bound.

11 The 214-state frontier: concrete obstructions and next targets

11.1 Why fewer source nodes did not automatically save a state

The attempted combination of direct transfer with MP-first command order has a valid abstract compiler and a better credit bound: only one protected head is needed at cutoff one. Some packed layouts have the same 201-node source size as ST215. That did not suffice for a physical 215-state allocation in the tested construction.

The old contextual implementation shares the CD1 and CD5 clear roles by inspecting a predecessor bit. Moving the direct-transfer clear under the MP-first branch changes that bit from one to zero, requiring an additional generic CD1 role unless a different contextual implementation is found. Early allocations passed source-prefix decoding but altered an arithmetic row used by CD8. Once the actually used 134/0 row was protected, the tested 201-node layouts became infeasible under that allocator's fixed-role policy. These rejected tables are not presented as valid candidates.

The lesson is specific: prefix decoding alone is insufficient, and source DAG cardinality is only one part of the physical allocation problem. The positive DR215 packet avoids inheriting a partial allocator mask as a runtime proof; it rederives all guarded physical cases for its actual table.

11.2 Fusing the transfer copy with its guard

A separate source-only improvement reduces the direct transfer body from 35 to 30 primitive occurrences. Clear T once, then pair every successful decrement of H in the cache-equals-one test and residual cleanup with an increment of T. At the completed pairs, $T + H$ equals the incoming cache. The zero case skips the halt with $T = H = 0$; the one case halts; the larger cases exhaust H into T and enter the unchanged Cantor unpair suffix. The latter restores the scratches and yields exactly the single-guard transfer contract.

Inserting an increment changes the loop's ancestor depth, so its comparison continue must be CC3 rather than CC2. The initial exploratory source missed that correction; only the corrected source is supplied. Its complete program has 4,727 occurrences but 208 distinct subtrees, against 4,732

and 201 in its parent. Thus it saves occurrences while losing sharing. The addendum includes the literal recipe, an all-natural source proof, dual-CFG checks and finite corroboration. There is no binary allocation, physical runtime certificate, or state-count claim for this source.

11.3 Sharing one selector between two marker mergers

The two new marker mergers suggest a genuine one-state target. M124 distinguishes current zero and M160 distinguishes current one. If their two internal selectors could be replaced by one shared selector, one physical label would become removable.

In the simplest three-step version, the marker-zero provider would need both actions

$$0 \mapsto 0R14, \qquad 1 \mapsto 1R99,$$

while the marker-one provider would need

$$0 \mapsto 0R62, \qquad 1 \mapsto 1R52.$$

No existing unchanged provider state supplies either required pair. Longer contextual provider blocks and temporary writes enlarge the search space, but must restore the complete tape and head position before rejoining the mapped old execution.

The supplied bounded shared-selector search tests both choices of retained helper label, temporary current-bit encodings, directions and writes of the two helper rows and two rewritten entry rows, and matching local contexts. Its radius-seven, fuel-97 primary-table census contains 46,895,104 specifications across the M124/M160 and earlier-M052/M160 families and has no survivors. This is a finite negative result for the stated local simulation model, not a proof that a shared selector or a 214-state machine is impossible. Longer simulations, different invariants, more simultaneous edits, or a changed arithmetic representation remain outside that conclusion.

The broader final census uses DR215 and all 200 disjoint pair combinations from a recorded inventory of 21 marker pairs: 105 combinations have opposite exceptional symbols and 95 have the same symbol. Temporary writes can encode pair identity even in the latter case. Both exceptional entry rows move left. Each family has

$$2 \cdot 4 \cdot (2 \cdot 2 \cdot 214)^2 = 5{,}861{,}888$$

parameter specifications, counting selector choice, temporary writes, and each selector row's write, direction and target. Across all families, 1,172,377,600 specifications were covered: 707,406,332 individually evaluated and 464,971,268 uniformly pruned, with no survivors. Pruning applies only when a failed path never reads the innermost variable row, so every remaining assignment to that row has the same failure. These are specifications, not necessarily distinct machines. Radius seven, fuel 97, a selected role inventory, an overapproximated context domain, and the prescribed edit shape limit the conclusion. The included C++17 sources, exact inventories and wrapper reproduce the census.

11.4 What would count as the next substantive advance

Three directions remain concrete enough to test:

1. A shared-selector construction with a proved invariant outside the present finite-window family. The displayed provider requirements make the missing interface explicit.
2. A direct MP-first encoding that implements its moved clear routine without the extra generic role. Its abstract compiler and protected-credit theorem are already available; its physical binding is the missing part.

3. A genuinely smaller refutation-equivalent numerical basis. Unlike the positive exchange family, this may change theorem sets, but it needs both primitive-recursive refutation directions. A classical equivalence argument that reuses an axiom being removed is not enough.

The measured failures help prioritize these targets. They establish no minimum-state theorem and do not justify calling 215 optimal. In particular, we have not supplied a smaller class theory, an arithmetic replacement for the set theory bundle, or a calibrated dual-sixth-row formalism; those would be different research projects with additional proof obligations.

12 Reproduction and artifact map

The archive contains this editable LaTeX source, its compiled PDF, the main candidate packet, marker family, formalism appendix, comparison corpus, licenses and provenance, and machine-readable hashes. The top-level README gives the exact final command lines. Positive candidate verification requires Python 3 with assertions enabled and no network or external repository checkout. Optional synthesis experiments may have additional dependencies, stated by their own scripts.

The default candidate replay reconstructs the literal source and both tables from the pinned parent; checks complete source/control differences; replays startup and the synthetic AX prefix; checks the dominated-row certificate; and audits the full symbolic case inventory. Its full-runtime option freshly rederives all 6,151 induction cases. The marker checker reconstructs all four tables and checks the finite premises of the all-tape theorem; a separately written auditor uses no builder imports. The graph audit compares rooted labeled transition graphs and reflections against the included inert corpus.

Runtime records and research receipts can retain original scratch paths as provenance. The positive reproduction commands use included relative inputs. They do not depend on those historical paths. SHA-256 manifests identify byte strings; they provide integrity checking, not a substitute for the mathematical arguments.

12.1 Attribution

The underlying architecture, native numerical theory, dominated-row calibration, register compiler, runtime induction engine, and direct-transfer and rho6 predecessors belong to the ProveIt contribution history. Its retained notices credit, within their respective scopes, ProveIt contributors, Elliot Glazer's compact theory work, Jeff's busy-beaver/Rocq reference, Andrew James Wade's compiler ancestry, BusyCoq, and the Coq Library of Undecidability. This contribution's exact composition, marker instances, exchange family, audits, and report were prepared by the research collaboration for Vladimir Reshetnikov. Independent agent reviews are internal review, not an external human endorsement.

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A Complete DR215 dense transition table

Each entry wDz means write w, move in direction D, then enter state z. The four groups of columns merely wrap the state list for printing. State zero starts on blank tape; 215 is the separate halt label. The machine file and explicit native/dense bijection are the authoritative machine-readable presentations.

q	0	1	q	0	1	q	0	1	q	0	1
0	1R91	1R0	54	0R22	1R54	108	0R148	1R183	162	0L108	1R170
1	0L1	1L2	55	0L11	0R50	109	0L204	0L94	163	0R144	1R169
2	0L3	1L2	56	0R60	0L16	110	0R123	1R103	164	0R193	1R28
3	0L5	1L2	57	0R69	1R70	111	0R206	1R180	165	0R158	1R141
4	1R6	0L4	58	0R61	1R71	112	0R162	1R178	166	0R175	1R175
5	0L9	1L2	59	0R54	1R59	113	0R146	1R144	167	0R178	1R121
6	0R6	0R10	60	0R48	0R25	114	0R157	1R197	168	0R143	1R162
7	0R83	1R187	61	0L28	1L72	115	0L130	0L1	169	0R75	1R60
8	0R14	1R13	62	0L116	1L30	116	0R38	1R72	170	0R143	1R87
9	0L15	1L2	63	0R73	1R63	117	0R61	1R131	171	0R103	1R95
10	0L16	0R10	64	0R96	1R148	118	0R61	1R132	172	0R138	1R83
11	0L26	1R128	65	1R181	1L53	119	0R70	0L16	173	0R172	1R75
12	0R19	1R20	66	0R142	1R143	120	0R14	1R131	174	0R20	1R37
13	0L88	1L31	67	0R76	1R215	121	0R21	1R133	175	0R140	1L147
14	0L11	0R22	68	0L193	1R31	122	0R145	1R175	176	0R96	1R51
15	0L23	1L2	69	0R40	1R62	123	0R135	1L20	177	0R165	1R51
16	1L1	0L1	70	0L88	1R77	124	0R129	1R177	178	0R17	1L130
17	0R24	0R32	71	0R55	1L78	125	0R201	1R168	179	0R76	1R115
18	0R198	1R197	72	0R79	1R80	126	0R75	1R173	180	0R105	1R123
19	0R28	1R29	73	1R0	1R63	127	0R17	1R212	181	0L39	1L109
20	0R30	1R31	74	0R74	1R81	128	0R75	1R134	182	0R168	1R12
21	0R24	0R59	75	0L143	1L139	129	0R12	1R139	183	0R148	1R114
22	0R33	1R22	76	0R84	1R85	130	0R30	1R69	184	0R11	1R97
23	0L34	1L2	77	0R88	1L131	131	0R38	1L123	185	0R189	1R98
24	0L35	0R0	78	0L107	1R56	132	0R151	0L16	186	0R144	1R195
25	0R36	1R25	79	0R48	0R33	133	0R152	0L16	187	0R154	1R188
26	0R127	1R37	80	0R90	0L41	134	0R158	1R137	188	0R75	1R182
27	0R38	0L39	81	0R194	1R174	135	0R24	0R47	189	0R64	1R7
28	0R40	1R41	82	0R95	1R12	136	0R138	1R177	190	0R148	1R126
29	0R42	1R43	83	0R87	1R96	137	0R191	1R93	191	0R153	1R111
30	0R44	1R45	84	0R203	1R21	138	0R95	1R156	192	0R150	1R122
31	0R21	1R46	85	0R115	1R99	139	0R172	1R163	193	0R89	1R87
32	0R47	1R32	86	0R148	1R112	140	0R12	1R156	194	0R214	1R115
33	0R32	1R33	87	0R101	1R102	141	0R87	1L163	195	0R202	1R108
34	0L49	1L2	88	0R79	1R78	142	0R87	1R155	196	0R148	1R166
35	1R35	1L1	89	0R100	1R171	143	0L150	1L160	197	0R196	1R157
36	0R50	1R36	90	0R104	1R105	144	0R83	1R87	198	0R138	1R136
37	0R13	1R52	91	0R65	0R0	145	0R156	1R12	199	0R192	1R18
38	0L11	0R36	92	0R27	1R106	146	0R82	1R141	200	0R87	1R205
39	0L4	1L53	93	0R179	0L204	147	0R161	1R165	201	0R160	1R190
40	0R24	0R25	94	1L109	1L94	148	0L66	1L139	202	0R125	1R149
41	0R13	0L16	95	0R19	1R110	149	0R146	1R199	203	0R44	1R99
42	0R55	1R56	96	0R60	1R60	150	0R156	1R113	204	0L4	0L204
43	0R57	1R58	97	0R164	1R154	151	0R96	1R96	205	0R193	1R138
44	0R24	0R54	98	0R66	1R186	152	0R48	0R32	206	0R161	1R207
45	0R60	1R61	99	0R104	0L204	153	0R68	1R200	207	0R208	1R179
46	0R62	0L16	100	0R116	1R117	154	0L207	1R95	208	0R141	1R209
47	0R25	1R47	101	0R71	1R58	155	0R165	1R176	209	0R142	1R210
48	0L35	0R63	102	0R57	1R118	156	0R19	1R167	210	0R211	1R86
49	0L74	1L2	103	0R21	1R119	157	0R170	1R162	211	0R185	1R162
50	0R16	1R50	104	0R21	0L10	158	0R156	1R95	212	0R184	1L72
51	0R145	1R141	105	0R61	1R41	159	0R124	1R161	213	0R92	1R107
52	0R67	1R68	106	0R120	1R8	160	0R96	1R159	214	0R213	1R127
53	0L39	1L53	107	0R100	1R121	161	0L7	1R140			